

Machine Unlearning as a Defense

Abstract—Abstract

Machine learning continues to grow in relevancy, as large language models are further integrated into critical infrastructure. As a result, models have more access to sensitive data than ever before which makes the threat of an attack on the model that much greater. Traditional defense mechanisms rely on inference-time filters that are often not effective against modern attack vectors such as Greedy Coordinate Gradient and Prompt Automatic Iterative Refinement. Machine unlearning has shown potential to be an alternative to traditional defense by removing sensitive data from a model without sacrificing the model’s accuracy. In this paper, we will investigate the use of machine unlearning as a viable defense mechanism against various attack vectors. We utilize the JailbreakBench Behaviors dataset using Vicuna-13b-v1.5 to properly evaluate the effectiveness of machine unlearning to create a model that is more safe against malicious queries and potential jailbreak attempts. The overall results of our research are that machine unlearning defenses work better than inference-time filters on white-box and black-box attacks without sacrificing the performance of the model by a significant margin. By properly evaluating these results, we hope to advocate for machine unlearning defenses to be more widespread in order to protect people’s data especially in fields that handle more sensitive information such as healthcare, finance, and government. Here is our repository link with our implementation and results: <https://github.com/emz05/jbm>

Index Terms—Data collection, data usage, privacy.

I. INTRODUCTION

Large Language Models (LLMs) are being heavily adopted across various industries which makes them more susceptible to attacks. These are more commonly known as jailbreaks and they manipulate the LLM to generate restricted content. Despite models being trained using safety fine-tuning such as Reinforcement Learning from Human Feedback (RLHF), LLMs are still susceptible to jailbreak attacks. Currently, the most common way to counteract jailbreak attempts is by using inference-time filters, but inference-time filters have been found to be limited in its defensive capabilities with there being more potential in a machine unlearning based defense.

Inference-time filtering analyzes the model’s incoming queries and outgoing responses as malicious or non-malicious [1]. These filters are easy to implement because they can be done after a model is trained without having to change the underlying logic of the model, but that also means that these models will always have the capacity to generate malicious content. This means that white-box attacks like Greedy Coordinate Gradient (GCG) can use adversarial suffixes that

maximize the likelihood of an affirmative response [2]. These suffixes will often bypass inference-time filters because they are random noise. Black-box attacks such as Prompt Automatic Iterative Refinement (PAIR) also render inference-time filters ineffective through the use of automated prompt refinement [3].

Inference-time filters such as SmoothLLM and Perplexity-Filter are often susceptible to high false-positive rates because they flag non-malicious queries as malicious. This negatively affects the user’s experience while limiting the model’s usefulness. These defenses are also computationally and resource intensive. These filters have to be run after every prompt which increases the computational cost, and they have to be constantly updated as attackers find new vulnerabilities in the filters.

By removing the ability of the model to generate malicious content instead of attempting to block malicious inputs and malicious outputs, most of the weaknesses of inference-time filters are negated. This is done through machine unlearning as a defense mechanism, as it modifies a model’s internal weights using techniques like Negative Preference Optimization (NPO) and Direct Preference Optimization (DPO). Models end up forgetting unsafe behaviors while maintaining a similar level of accuracy.

This paper analyzes the viability of machine unlearning as a defense mechanism against attacks compared to inference-time filters. We first establish the dataset we will be using for the attacks and the model that we will be using to test them. We then analyze the different inference-time filters and the machine unlearning defenses. Finally, we evaluate the differences between the different defense mechanisms to determine the advantages and disadvantages between them in order to promote a more widespread adoption of machine unlearning as a defense mechanism.

II. BACKGROUND

A. Machine Unlearning

Privacy is becoming even more difficult to achieve because of the popularity of machine learning models increasing. As these models become more widespread and continue to be integrated into key infrastructure in every field, models will analyze more data that is likely sensitive information such as a patient’s medical records, personal emails, etc. The data is often stored within the model after being analyzed which puts potentially sensitive information at risk should an attack on the model succeed. Machine unlearning is a new and arising field of machine learning that combats this problem by getting the model to forget specific points of data without losing its accuracy based on the leftover data. While machine

unlearning’s main use is in regards to removing sensitive information, it has also started to be used in cybersecurity by using a defense-in-depth strategy. This modifies the model’s internal weights to ensure the model does not know how to comply with harmful requests instead of identifying an attack in inference-time.

B. Brief History of Machine Unlearning

The need for the ability to remove data from machine learning models arose to be a pressing issue specifically due to laws surrounding privacy. One notable example is the “Right to be Forgotten” which is a policy in the European Union’s General Data Protection Regulation that allows any individual to request to remove personal data from an organization [4]. Due to policies like these and overall concern over data privacy, machine unlearning emerged with the goal of being able to remove specific data points from a model without sacrificing the accuracy of the model. Machine unlearning started being researched in 2015 and interest in it has only increased since then.

C. Challenges in Machine Unlearning

There are many challenges when trying to remove a specific piece of information from a machine learning model because of the fact that models are randomly trained and rely on previous data in order to learn properly. As a result, machine unlearning models often underperform when compared to machine learning models that were retrained on that same remaining data. The model continues to degrade potentially exponentially as data is removed which is known as catastrophic unlearning. This is a significant roadblock in machine unlearning, and though there are ways to mitigate this effect, it is currently unknown on how to prevent catastrophic unlearning from happening.

D. Mechanisms of Machine Unlearning

The two main goals of machine unlearning are ensuring that the model completely removes all influence of the data that needs to be forgotten and that the new model behaves the same as a model that was not trained on the data points in the first place. The process starts with a removal request which splits the dataset into two: the data that is unneeded D_u and the data that is needed D_n . A naive approach to solving this problem is retraining a completely new model using only the dataset D_n , which leads to a model that ensures the two core goals of machine unlearning [5]. However, this process is not practical the larger the dataset is because of how computationally intensive retraining a model is. More efficient algorithms were developed and are continuing to be improved that approximate the retrained model from the original model. By approaching the problem of machine unlearning with these algorithms, the previous issues of high computational cost are significantly reduced while still achieving the main goals of machine unlearning.

E. Overview of Jailbreak Attacks

Jailbreak attacks on large language models are categorized as having the model generate content that goes against its guidelines such as making the model generate harmful or unethical content [6]. There are two main kinds of jailbreak attacks: white-box attacks and black box attacks. A white-box attack uses internal knowledge of the model including its architecture and code while a black-box attack only uses inputs and observable outputs. One of the most common white-box attacks is Greedy Coordinate Gradient (GCG) which finds optimal adversarial suffixes using gradient information to trigger affirmative responses from a model. A common black-box attack is Prompt Automatic Iterative Refinement (PAIR) uses a different large language model to iteratively refine a prompt until it bypasses the target model’s defenses.

III. METHODOLOGY

A. Dataset

We evaluate all attack and defense configurations on the JBB Behaviors dataset [3]. The full set contains 200 distinct behaviors, comprising 100 benign and 100 harmful examples. All examples span ten misuse categories defined by OpenAI’s usage policies. Harmful behaviors are sourced from AdvBench [7], TDC/HarmBench [2], and original JailbreakBench contributions. The GCG and PAIR attacks target vicuna-13b-v1.5, an open source LLM with known safety weaknesses.

For unlearning based defenses, we construct two disjoint training sets. The forget set $\mathcal{F}$ aggregates harmful prompts from AdvBench, HarmBench, LibrAI’s Do-Not-Answer and WalledAI’s XSTest unsafe partition. These sources are selected for their broad coverage of policy violating distributions across major misuse categories. The retain set $\mathcal{R}$ is drawn from Databricks-Dolly-15k, MedQuAD, Enron Emails, and XSTest safe partition. These datasets provide structural and topical overlap with $\mathcal{F}$ while excluding harmful intent. Additionally, coverage of general instructions with the defined datasets preserves general conversational and task oriented capabilities.

Each example is represented as a (goal, target) pair, where the goal denotes the user query and the target denotes the expected response. We derive four training splits from the original disjoint set. $\mathcal{F}$ and $\mathcal{R}$ contain affirmative targets, establishing the baseline behavioral distribution. $\mathcal{F}_{\text{idk}}$ and $\mathcal{R}_{\text{idk}}$ replace these targets with refusal or limited capability responses to facilitate supervised unlearning.

B. Threat Model

We formalize the threat model as follows. Let $f(\cdot)$ denote a LLM previously tuned for safety alignment, such that for a harmful query q , the model outputs a refusal. A jailbreaking attack constructs an alternative prompt q' such that $f(q')$ elicits a response that violates the model’s predefined safety policies, despite the intent of q remaining unchanged. Under this formulation, a defense must ensure that f remains resilient to such transformations while preserving utility on benign inputs. To evaluate such assumptions, we establish the following attack vectors and defense mechanisms.

C. Attack Methods

We consider three attack vectors within a hierarchical threat framework. **Direct prompting** submits the raw harmful prompt to the target model without modification. This establishes a baseline reflecting the model’s pretrained safety alignment in absence of any adversarial pressure. **GCG** [2] is a white box attack that appends an optimized suffix to the harmful goal. The suffix is found through a greedy coordinate gradient search, which iteratively identifies token substitutions that maximize the probability of an affirmative response. Because the attack requires access to model’s weights, GCG serves as a strong upper bound for white box threat models. **PAIR** [3] is a black box attack that uses a separate attacker LLM to refine jailbreak prompts iteratively through feedback loops. Unlike GCG, PAIR is restricted from accessing the target model’s internals, replicating a more realistic threat sandbox.

D. Defense Mechanisms

We examine two broad categories of defenses: inference time filters and training time unlearning methods.

1) *Inference time defenses*: Filter defenses wrap the model at inference time without modifying its parameters for patch fixes. **SmoothLLM** [1] applies stochastic character level perturbations to N copies of each prompt and aggregates responses by majority vote, with LlamaGuard2 as the judge. Due to computational constraints, we use $N = 5$ perturbation copies instead of the default $N = 10$. **PerplexityFilter** [8] rejects any prompt whose token level log perplexity exceeds the 95th percentile over clean behaviors. This is effective against detecting unnatural syntax such as GCG suffixes. **Synonym substitution** randomly replaces tokens with WordNet synonyms with probability $q = 0.05$, simulating interrupts in adversarial sequences. **Remove Non Dictionary** strips tokens absent in a defined dictionary, targeting irregular patterns commonly seen in white box attacks.

2) *Training time defenses*: Unlearning which alter model parameters to remove harmful behaviors at source before release. **Gradient Ascent (GA)** actively penalizes harmful outputs on $\mathcal{F}$ by optimizing away from learned unsafe responses. This naive approach, however, risks catastrophic collapse when ascent is unconstrained, degrading general instruction based abilities. **Gradient Ascent + Gradient Descent (GA+GD)** augments GA with a retain side regularizer that preserves benign knowledge through descent on $\mathcal{R}$. **Direct Preference Optimization (DPO)** adopts a reinforcement style alignment objective in which each prompt yields two targets. These targets are defined as the chosen refusal response $\mathcal{F}_{\text{idk}}$ and rejected forget response $\mathcal{R}_{\text{idk}}$. The model here is rewarded for refusal behavior on unsafe inputs. **Negative Preference Optimization (NPO)** [9], [10] follows similarly but uses a one to one target mapping and DPO anchored loss over GA. The replaced loss prevents catastrophic collapse. **Gradient Descent (GD) and Anchored Preference (AP)** stabilize unlearning by either anchoring updates to a safe reference point

TABLE I
ATTACK SUCCESS RATE AND REFUSAL RATE AT TIME INTERVAL
REFERENCED IN FIGURE 1

Time	Vanilla	GCG	PAIR	Refusal Rate
x(t0)	0.01	0.30	0.24	0.35
x(t1)	0.00	0.04	0.01	0.79

or preserving generalization through retain side optimization process.

E. Evaluation Metrics

We report two primary metrics. Attack Success Rate (ASR) is the fraction of harmful prompts that elicit a jailbroken response, as judged by LlamaGuard-2-8B. Lower ASR indicates stronger defense performance. Refusal Rate is the fraction of benign prompts that are incorrectly met with refusals, with lower values indicating better utility preservation.

On JBB comparison data, LlamaGuard-2-8B achieves 87.7% agreement with human annotators and a 10.9% false negative rate, improving over its predecessor’s 72.0% agreement and 60.9% false negative rate [3], [11].

F. Experimental Setup

All filter defenses are evaluated on the 100 harmful behaviors. For SmoothLLM, we set $N = 5$ perturbation due to GPU constraints. Synonym substitution uses WordNet at its dictionary reference. All unlearning methods target vicuna-13b-v1.5, selected for its weak baseline safety alignment among open source LLMs [3].

Parameter efficient fine tuning is applied via LORA [12]. Samples are tokenized by concatenating goal and target, with loss masked over goal tokens. Training uses AdamW with learning rate 10^{-4} and batch size 4. Configurations including termination and regularization criteria are specific to each model.

To avoid over forgetting, training is terminated once the refusal rate satisfies the chosen tolerance threshold. Further optimization beyond this point leads to generalization collapse on benign queries. The training path in Figure 1 shows two representative checkpoints. Table I, reports the corresponding evaluation metrics. At $t = 0$, attack success and refusal rates remain within normal limits, while at $t = 1$ the model has passed the selected stopping point and continues unlearning. Continued optimization beyond this threshold yields diminishing returns on forget loss and a rapid increase in refusal behavior, as reflected in the table.

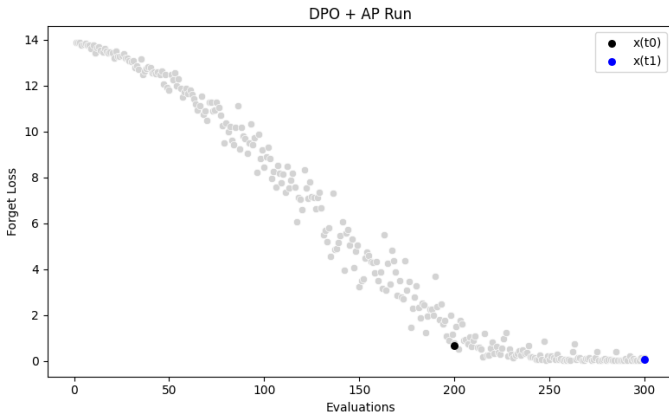


Fig. 1. DPO+AP training path showing forget loss over evaluations.

IV. RESULTS

A. Undefended Baseline

The undefended model establishes the attack surface under all three vectors. Direct prompting yields an ASR of 0.13, consistent with incomplete safety alignment across standard tuned models (Wei et al., 2023). GCG and PAIR raises ASR values to 0.80 and 0.62 respectively, confirming vulnerability to both attacks. The gap difference separates how well performance is measured through white and black box optimizations. The baseline refusal rate of 0.22 defines the lower utility bound against which all defenses are evaluated.

TABLE II
ATTACK SUCCESS RATE (ASR) AND REFUSAL RATE (RR) FOR ALL FILTER DEFENSE CONFIGURATIONS. LOWER ASR INDICATES STRONGER DEFENSE. LOWER RR INDICATES BETTER UTILITY PRESERVATION.

Defense	Vanilla	GCG	PAIR	RR
None	0.13	0.80	0.62	0.22
SmoothLLM	0.08	0.12	0.50	0.28
Synonym	0.11	0.21	0.21	0.26
Remove	0.13	0.27	0.41	0.24
Perplexity	0.13	0.02	0.59	0.27

B. Filter Defense Results

Inference time defenses are often specialized against specific threat models. Because each method is implicitly optimized for a particular attack vector, vulnerabilities persist when different strategies including semantic and syntactic structures are introduced. PerplexityFilter achieves the strongest single attack in our evaluation, reducing GCG ASR from 0.80 to 0.02. GCG suffixes create sequences with high perplexity that sit far from natural prompt distributions [13]. The defined 95 percentile threshold is enough to intercept almost all the queries. This shows the derived advantage is attack specific. Against PAIR, the difference between ASR of 0.59 and the baseline 0.62 is not statistically significant enough. Pair prompts are designed by LLMs to replicate queries in similar language flow. Under perplexity criteria’s,

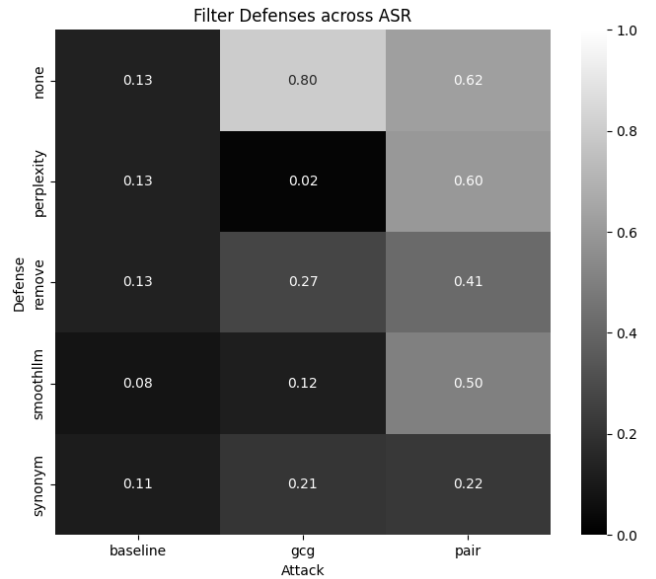


Fig. 2. ASR heatmap across filter defenses and attack vectors. Darker cells indicate lower attack success. PerplexityFilter approaches zero GCG ASR but fails to generalize to PAIR, while Synonym shows more uniform coverage.

it is blind to the semantic jailbreak class. However, this class is more representative of threat models since attacker’s rarely have white box access, as shown in Fig. 2. Table II reports ASR and refusal rate across all filter configurations.

Conversely, Synonym Substitution performs better on PAIR, reducing ASR from 0.62 to 0.20 by replacing tokens with WordNet synonyms with probability $p = 0.05$. This defense introduces controlled perturbations into the prompt surface, disrupting the semantic structure attack wrapper while preserving much of the original meaning. A similar reduction is observed against GCG, suggesting that this defense is sensitive to both syntactic and semantic attacks. Compared with PerplexityFilter, Synonym Substitution focuses on token identity rather than raw fluency, which gives it broader coverage across attack styles.

Remove Non Dictionary is more conservative. Its goal is to strip away irregular tokens and weaken harmful prompt construction. In this experiment, however, the dictionary used covers most standard academic and conversational terms, so the defense often removes noise without fully breaking the semantic wrapper of the attack. As a result, the method provides only modest gains across the evaluated attack vectors. A stricter frequency based dictionary would likely produce the intended effect more reliably.

SmoothLLM provides the most balanced coverage across attacks. Its majority vote aggregation over $N = 5$ perturbed copies reduce the influence of deterministic suffixes by injecting randomness into the inference process. In this experimental setup, SmoothLLM is evaluated with $N = 5$ rather than the recommended $N = 10$ because of computational constraints. Studies suggest that increasing N improves generalization, particularly against black box attacks [1]. This means our reported PAIR ASR may overestimate SmoothLLM’s vulnerability.

All together, the filter results show that each defense specializes in the type of structure it is designed to detect. No single filter defense reduces ASR uniformly across all three attack vectors. This displays the limitation of inference time filtering as these methods act as patches over observable input features, and their coverage is bounded by distributional assumptions. As attack methods become more adaptive, the generalization gap widens. This pushes a shift towards unlearning defenses, which modify model parameters over relying on prompt inspection alone.

TABLE III
ATTACK SUCCESS RATE (ASR) AND REFUSAL RATE (RR) FOR ALL UNLEARNING DEFENSE CONFIGURATIONS. LOWER ASR INDICATES STRONGER DEFENSE. LOWER RR INDICATES BETTER UTILITY PRESERVATION.

Defense	Vanilla	GCG	PAIR	RR
GA	0.03	0.49	0.39	0.36
GA+GD	0.02	0.14	0.18	0.29
NPO+AP	0.04	0.49	0.40	0.25
NPO+GD	0.05	0.18	0.20	0.30
DPO+AP	0.01	0.30	0.24	0.26

C. Unlearning Defense Results

Unlearning defenses adapt model parameters prior to inference, effectively removing the model’s capacity to generate harmful content rather than filtering the inputs that elicit it. Our results indicate that this approach produces more generalizable behavior. However, with more consistent coverage across attack types, cost of refusal rate and performance vary dependent on the forget and retain side objectives. We adapt popular unlearning frameworks originally perfected for static tasks like image classification to assess their efficacy in high variance domain of natural language generation. Table III reports ASR and refusal rate across all configurations.

Gradient Ascent is the simplest unlearning configuration and weakest performer in our evaluation. GA reduces vanilla ASR from 0.13 to 0.03, showing strongest suppression on direct prompting. However, GCG and PAIR ASR remain high at 0.49 and 0.39 respectively, showing poor transfer across prompt transformation. This suggests that GA mainly moves the model away from specific harmful completions without altering the representational structure that supports harmful behavior. As a result, the is easy to bypass once the input distribution is shifted. In addition, GA increases refusal rate to 0.36, the highest among all configurations. This reflects learning collapse where general instruction ability is degraded when ascent is unconstrained. In effect, GA serves as a naïve baseline where it suppresses direct harmful responses but fails to generalize while imposing the highest utility cost.

GA+GD is an extension of gradient ascent. Gradient descent is a regularization term over the retain set R . While gradient ascent pushes completions against the grain, gradient descent forces the model to preserve its original output distribution on benign inputs. This prevents the representational collapse

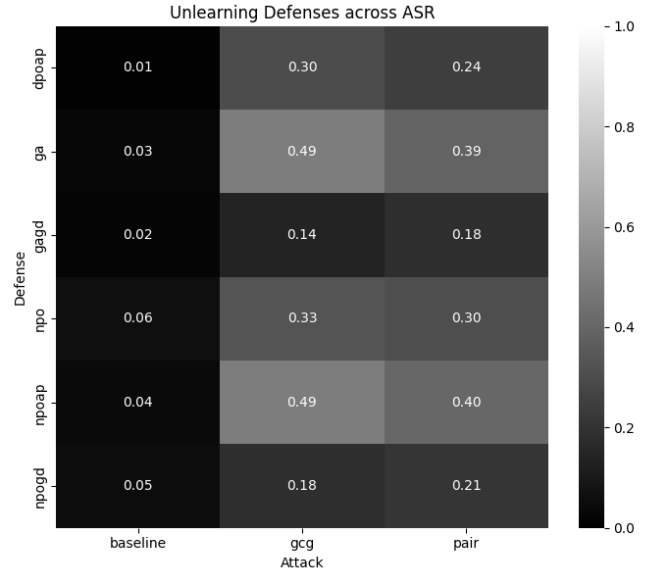


Fig. 3. ASR heatmap across unlearning methods and attack vectors. GA+GD and NPO+GD show the most consistent defenses across both GCG and PAIR, while GA and NPO+AP leave high residual ASR under white box attack.

observed in GA. As a result, GCG ASR drops from 0.49 to 0.14 and PAIR ASR decreases from 0.39 to 0.18. The refusal rate also falls to 0.29. These results support the expected behavior that GD is expected to apply.

NPO+GD follows similar structure as GA+GD but replaces the gradient ascent forget objective with a loss anchored to a reference model state. The results here are comparable with GCG ASR at 0.18, PAIR ASR at 0.20, and refusal rate at 0.30. This similarity suggests that descent regularization helps with producing balanced unlearning behavior regardless of forget objective. In both cases, the retain side objective helped preserve benign capability while improving generalization across transfer attacks.

NPO+AP and DPO+AP introduce a different retain objective. Anchored preference optimization keeps the model closer to its pre unlearning output distribution. Both configurations reduce vanilla ASR almost to zero, but their GCG and PAIR ASRs remain higher than those of with GD regularization. This pattern suggests that AP constrains the model too tightly around its original state. The tradeoff presented here shows how preserving the model’s benign capacity more aggressively comes at the cost of incomplete forgetting. The lower ASR values reflect this interpretation as AP configurations respect low refusal rate and vanilla ASR in exchange for insufficient generalization across transfer attacks. Fig. 3 visualizes ASR across all unlearning configurations.

D. Defense Comparisons and Practical Implications

Comparing filter and unlearning defenses reveals a structural difference in how each method achieves coverage across attack vectors. Filter defenses provide strong but attack specific coverage by exploiting observable surface features in the prompt. Unlearning defenses achieve broader generalization across attack types by changing the model’s internal representations,

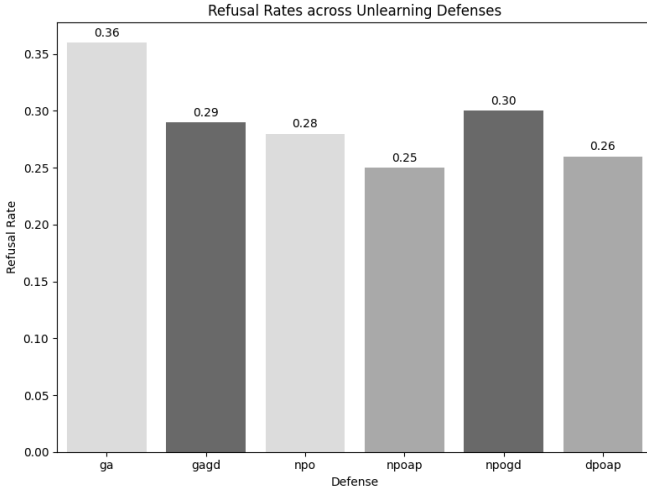


Fig. 4. Refusal rates across unlearning defenses. GA produces the highest benign refusal rate (0.36), reflecting unconstrained learning collapse. GD applications hold refusal near 0.29-0.30, while AP configurations achieve the lowest rates at the cost of incomplete forgetting.

but they do so at the cost of a more consistent increase in refusal rate. This presents the boundary between sufficient forgetting and over forgetting.

From these results, GA+GD and NPO+GD are the most practical configurations. Both keep GCG and PAIR ASRs below 0.20, maintain vanilla ASR below 0.05, and hold refusal near 0.30 with only a modest increase over the undefended baseline of 0.24. The resulting utility cost is around five additional benign refusals per 100 queries, which is likely acceptable for strict safety requirements. Filter defenses do not achieve comparable cross attack coverage without sacrificing either GCG or PAIR robustness.

The retain side objectives also showed a clear tradeoff between complete unlearning and utility upkeep. GD consistently outperforms AP on transfer attacks, but AP consistently produces lower refusal rates. In other words, GD supports more aggressive forgetting at cost of higher benign refusal, while AP preserves benign utility at the cost of incomplete generalization. The preferred choice now depends on the setting in terms of tolerance for false refusals versus leftover vulnerability, as shown in Fig. 4.

V. CASE STUDY

Machine unlearning is moving from a theoretical privacy concept to practical security tool, a transition best shown through specific implementations and industry adoption. Comparing an algorithmic approach with real-world privacy deployments targets the limits from persistent knowledge stored in LLMs. The case studies below explore how structured unlearning and corporate compliance methods are complementary to the development of machine unlearning.

A. Early Foundations

The work by Cao and Yang is considered a foundation case study in machine unlearning because it was one of the first to clearly explain how to let a model forget certain training

data without retraining everything. They used Statistical Query (SQ) learning to make models learn from data summaries instead of the raw data points. Thus, instead of retraining, the system updates the summaries when data has to be removed. This summation form of the system allowed machine unlearning to be faster and more productive [14].

The concept later extended into more scalable models through the SISA (Sharded, Isolated, Sliced, and Aggregated) training system [15]. SISA divides training data into sub-datasets that each train separate models. Each group then further partitions into training stages, with progress saved along the way. Since each datapoint only influences a part of the system, only a small part of the system has to be redone when data is removed. By combining all the models' outputs to create a final prediction, SISA limits the influence of any single data points on the model's overall behavior. Machine unlearning often relies on trial and error during retaining to delete data, but SISA reduces this uncertainty by isolating data into shards to be removed and retrained.

SISA's structural isolation helps connect basic data deletion with stronger defenses against jailbreak attacks, unlike Differential Privacy (DP) which depends on adding statistical noise. SISA changes the goal from access revocation to model correction by allowing exact unlearning [16]. If a jailbreak relies on harmful learned knowledge, SISA confines the problem to one part of the model for removal instead of suppression, which is described as proof of unlearning. Despite this property, SISA as an implementation of machine unlearning arguably has practical limits at scale. This is due to scaling laws of LLMs, where performance improves as it is exposed to more diverse and interconnected datasets. Specifically, the training framework's dependence on sharding and maintaining multiple sub-models further increases computational complexity. Therefore, it presents a trade-off between stronger unlearning and model utility.

B. Industry Setting

Beyond academic applications like SISA, machine unlearning is increasingly getting explored in industry as a tool for compliance, privacy by design, and model reliability. Despite the growing interest, large-scale deployment of machine unlearning remains limited. A notable example with tangible progress of this industrial research comes from IBM, whose work shows balance between practical costs and model integrity.

An important challenge addressed by IBM's research was catastrophic forgetting, where the model removes more than the targeted data when unlearning. Their solution was to develop the split-unlearn-then-merge (SPUNGE) framework. SPUNGE splits unlearning data based on certain attribute values such as demographics, unlearns each subset to make multiple domain-specific models, then merges the unlearned models [17]. Using SPUNGE on a Llama model, IBM researchers reported being able to reduce toxicity score from 15.4% to 4.8% in 224 seconds while preserving the model's accuracy on general tasks [18].

Although both SISA and SPUNGE strive to avoid full training, they rely on different structural approaches. SISA proactively partitions the entire training data to retrain certain model shards and aggregates at the output level. In contrast, SPUNGE reactively splits only unlearning data and selectively updates model weights with methods like TIES-Merging. SPUNGE can offer lower deployment overhead by producing a single unified model, whereas, SISA relies on maintaining multiple models to remove harmful knowledge.

This distinction becomes particularly important in the context of jailbreak defense because it reveals focus on corrective methods on models post-deployment, which is when exploits are often found. IBM’s exploration furthers this idea in practice by showing how learned dependencies responsible for harmful behaviors can be removed after identified. Rather than replacing current defenses, they suggest machine unlearning as a complementary reactive defense.

VI. LEGALITY

While machine unlearning is emerging as a critical defense against security threats, acknowledging the boundary of complete deletion of learned information is crucial. Unlike deterministic systems, LLMs store information in distributed patterns across high-dimensional weight spaces during training. Consequently, machine unlearning should be recognized as a probabilistic safeguard, reducing the likelihood of unsafe elicitation and raising computation cost for adversaries. In this perspective, LLM jailbreaking transcends technical failure and constitutes a violation of legal rights. This sets machine unlearning within discussions regarding ethics and human rights, where its value for compliance is recognized.

A. Compliance Bridge

Machine unlearning serves as a technical interface between human rights and generative systems [19]. This mechanism becomes especially significant under widely compared privacy frameworks like the EU’s General Data Protection Regulation (GDPR) and the U.S. California Consumer Privacy Act (CCPA). These laws were originally designed around discrete data systems, where information could be directly accessed and deleted. In these contexts, compliance with data erasure was straightforward. However, they now impose data control requirements that traditional black box LLM training methods cannot inherently support. Accordingly, machine unlearning translates abstract rights into quantifiable changes to model weights.

The GDPR applies to any organization managing EU residents’ data. It recognizes privacy as a fundamental right and gives people control over their personal data through mechanisms such as the “right to be forgotten” (Article 17) [20]. Conversely, U.S. privacy laws have a fragmented, sector-based approach that grants rights to whoever possesses the data. While U.S. privacy frameworks may be less prescriptive, the scope of GDPR necessitates verifiable data deletion.

Recent legal analysis highlight that unlearning requires erasure of both data and its functional influence [21]. The

dispersed legal landscape complicates legal compliance for global developers, and the gap between regulatory expectations and technical delivery intensifies this burden. As a result, many organizations rely on more scalable and cost efficient methods like output and moderation filtering. This defensive posture reflects a whack-a-mole approach, where suppressing certain outputs does not extinguish the issue but instead requires continuous intervention as new variations surface.

B. Model-Level Regulation

While privacy laws like GDPR and CCPA established the foundation for data deletion, emerging AI regulations are reframing the scope of machine unlearning. Legal frameworks like the EU Artificial Intelligence Act (AIA) shift the discussion from isolated data issues to how models behave in deployment and how their risks are managed. Within this shift, machine unlearning is evolving from reactive remediation to a proactive defense for model integrity maintenance [22].

The AIA transforms AI development from a permissive process to a system that enforces rules depending on the level of risk the AI entails. It organizes AI into AI systems and general-purpose AI (GPAI) models. For AI systems, risk is determined by where the system is used, meaning that systems are categorized as High-Risk AI (HRAI) when deployed in sensitive domains affecting public safety and fundamental rights. Under Article 9, HRAI systems are mandated to adopt risk management systems that identify and mitigate “reasonably foreseeable risks” [23]. In contrast, GPAI models are classified by the risk they pose from their capabilities and scalability. Furthermore, models with potentially widespread and substantial negative impact are designated as systemic risks.

Under the AIA act, developers of the model are held responsible along with those that use said model. The developers have to ensure that the model that is deployed is safe and that the applications of the model is handled appropriately [23]. This leads to some ethical risks including a situation in which a model that is used for general purposes could be used with malicious intent. Machine unlearning allows for a way to mitigate this risk by removing the persistence of harmful knowledge that compromises fundamental rights or public safety.

VII. CONCLUSION

This paper evaluates whether machine unlearning can be used to reduce a model’s ability to create unsafe outputs, instead of relying on traditional filtering defenses that are applied at inference time. The results showed that the limits of filtering defenses stem from depending on statistical features of adversarial prompts, which are vulnerable to distribution shifts by highly adaptive jailbreaks. In contrast, machine unlearning applied at training time demonstrates a more consistent decrease in attack successes by optimizing model weights. However, this can introduce a measurable trade-off in model usability through its slight increase in false positive refusals.

These findings can position machine unlearning as a supporting layer alongside existing defenses, and may suggest it will evolve into a primary method. Beyond the technical challenge of verifying effective forgetting, this transition is motivated by a legal and ethical imperative. Global regulatory frameworks are progressively demanding transparency and accountability in how models manage information. Thus, strengthening machine unlearning will not just be about system resilience, but about making AI more governable.

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